Area and volume — Part 2 answer key

#	ID	Answer	Difficulty
2.1	d341e9e5-5270-4fed-85ea-10d666d20e91	7.5, 15/2	Medium
2.2	0baecf54-d44d-4084-ba8f-a8fb9f7596bf	D	Medium
2.3	f60d8609-7495-41f1-a588-c1fdce04101d	C	Medium
2.4	d1c64540-1ef9-4449-9ccf-d158b23af654	D	Medium
2.5	0103767c-8ef7-4ec8-a9fb-3a3fa9a5469f	A	Medium
2.6	eee80adc-71a4-4db9-8c03-1fddc462909d	C	Medium
2.7	5b086c61-e308-4d87-b595-7b21a6d05163	D	Medium
2.8	d0ab21dc-5c27-4da2-a45b-fa400982d6bc	A	Medium
2.9	038021-DC	C	Medium
2.10	b7b7672e-ce42-4e19-9094-c2ba6966c496	D	Medium
2.11	9d90d9a2-0084-4cae-be11-73b040f11c9e	D	Medium
2.12	cc551413-74ed-4f97-8552-a6ef542a8584	B	Medium
2.13	fc7a142b-274d-4f6f-ab55-5f7708568898	D	Medium
2.14	b609067c-22bb-404f-a813-9e197c6b487c	D	Medium
2.15	63912a12-3cfc-4144-bd81-5557667cbb81	D	Medium
2.16	df0a6944-eb80-45dd-890a-e82089329b8e	14	Medium
2.17	020533-DC	?	Medium
2.18	027063-DC	A	Medium
2.19	b6420961-9ecc-40b1-9203-ca168e783426	C	Medium
2.20	aa19a0c6-76f9-4d66-b105-5e444ac3faec	D	Hard

2.21	205ab66a-8e60-4a19-9cbb-9c2754ce529f	D	Hard
2.22	c19a1158-3fde-4af0-be5d-79e3722ddf7e	2592/5, 518.4	Hard
2.23	25288de3-8156-4952-8e45-10506cfad636	D	Hard
2.24	395e8a55-7af9-456e-a433-3d4a89ade971	2216	Hard
2.25	d6c651bd-349f-4a16-89ba-e8bb62e72076	A	Hard
2.26	131e158b-3eee-4f01-b713-1985fda38651	A	Hard
2.27	3dc2565f-6841-467d-aeb0-c0ac79807ad9	50	Hard
2.28	70b95bcc-29ac-4b72-909d-c64fad3c9c1d	B	Hard
2.29	9a0d902b-5ffd-4b9f-99b5-e223fb690c35	4	Hard
2.30	05721-DC	C	Hard
2.31	ca49552d-a377-4804-b20d-7b730e15bb99	1260	Hard
2.32	adc9a053-ace5-4475-8499-b52461f5f6e1	594	Hard
2.33	025459-DC	B	Hard
2.34	9657587b-66c2-4743-91d3-22e3bf8ab94e	135/8, 16.87, 16.88	Hard
2.35	a5e0e74e-362f-4f21-aade-6a5d1b2518e0	C	Hard
2.36	243579cd-f8bb-4573-bce8-ea57314f3f36	66	Hard
2.37	c682f49c-ee6e-439a-a3c9-64f6aab47176	B	Hard
2.38	95dd10f0-9b55-4202-a756-91f1ede906ac	B	Hard
2.39	bca754b0-4853-461c-a71b-c465e54cbb57	12	Hard
2.40	3b48c674-430c-4197-b876-ffb5199bc927	1681	Hard
2.41	518073a0-4d7c-48ab-af96-3d1d19b33661	C	Hard
2.42	01daee35-6a84-4baf-8ed5-3ef4ff55b251	A	Hard
2.43	c1c7e4d7-d6dc-4e54-8b49-cbd88eab7153	C	Hard

2.44	06496-DC	D	Hard
2.45	e017d068-1e14-45a7-8981-963bc0f7fc9a	8	Hard
2.46	31c28a2a-1001-4c2c-b27f-aa2b2380c77d	36504	Hard
2.47	032655-DC	A	Hard
2.48	1f5958dc-ffc1-426c-8358-d8fbb4354de4	C	Hard
2.49	555b2745-a82f-4231-a6df-c31085a80638	27556	Hard
2.50	bb20b967-73fe-4c5e-97f8-e5887afb9aab	A	Hard
2.51	8f07857d-c0af-4793-9305-0e414671d257	24.5, 49/2	Hard
2.52	b44ba6ac-80e2-4dec-ab99-a08180e2db81	5800	Hard
2.53	038033-DC	?	Hard
2.54	5c9efc7d-a1db-4027-b9b5-a10ca2abedcd	B	Hard
2.55	64a7c0bf-065d-47a5-a938-4a1fabad6933	D	Hard

Rationales

▼ 2.1 — 7.5, 15/2

The correct answer is $\frac{15}{2}$. The area, A, of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. In the right triangle shown, the length of the base of the triangle is 5 inches, and the height is 3 inches. It follows that $b = 5$ and $h = 3$. Substituting 5 for b and 3 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}53$, which is equivalent to $A = \frac{1}{2}15$, or $A = \frac{15}{2}$. Therefore, the area of the triangle, in square inches, is $\frac{15}{2}$. Note that 15/2 and 7.5 are examples of ways to enter a correct answer.

▼ 2.2 — D

Choice D is correct. The volume, V, of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius of the circular base and h is the height of the cylinder. It's given that the cylinder has a height of 8 meters and a base with a radius of 12 meters. Substituting 12 for r and 8 for h in $V = \pi r^2 h$ yields $V = \pi(12)^2 8$, or $V = 1{,}152\pi$. Therefore, the volume, in m^3, of the cylinder is $1{,}152\pi$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the volume, in m^3, of a cylinder with a radius of 8 meters and a height of 12 meters.

▼ 2.3 — C

Choice C is correct. The volume, V, of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of 22 centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or 11, centimeters. Therefore, $r = 11$. Substituting 11 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi 11^2 6$, which is equivalent to $V = \pi 1216$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is 726π.

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not 22, centimeters and a height of 6 centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not 22, centimeters and a height of 6 centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of 44, not 22, centimeters and a height of 6 centimeters.

▼ 2.4 — D

Choice D is correct. The area of a square is given by s^2, where s is the length of each side of the square. It's given that the length of each side of a square is 94 cm. It follows that the area, in cm^2, of the square is 94^2, or $94 \cdot 94$. Therefore, the expression that gives the area, in cm^2, of the square is $94 \cdot 94$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression gives the perimeter, in cm, of the square.

▼ 2.5 — A

Choice A is correct. The area, A, of a square is given by the equation $A = s^2$, where s is the length of one side of the square. Let x represent the length, in feet, of one side of square N. It's given that the area of square N is 49 square feet; therefore, it follows that $49 = x^2$. It's also given that the length of one side of square M is 6 times the length of one side of square N. Therefore, the length of one side of square M can be represented by the expression $6x$. Substituting $6x$ for s in the equation $A = s^2$ yields $A = (6x)^2$, or $A = 36x^2$. Since $49 = x^2$, substituting 49 for x^2 in the equation $A = 36x^2$ yields $A = 36(49)$, or $A = 1{,}764$. Therefore, the area, in square feet, of square M is $1{,}764$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.6 — C

Choice C is correct. The base of a cylinder is a circle with a diameter equal to the diameter of the cylinder. The volume, V, of a cylinder can be found by multiplying the area of the circular base, A, by the height of the cylinder, h, or $V = Ah$. The area of a circle can be found using the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the diameter of the cylinder is 8 inches. Thus, the radius of this circle is 4 inches. Therefore, the area of the circular base of the cylinder is $A = \pi 4^2$, or 16π square inches. It's given that the height h of the cylinder is 12 inches. Substituting 16π for A and 12 for h in the formula $V = Ah$ gives $V = 16\pi 12$, or 192π cubic inches.

Choice A is incorrect. This is the area of the circular base of the cylinder.

Choice B is incorrect and may result from using 8, instead of 16, as the value of r^2 in the formula for the area of a circle.

Choice D is incorrect and may result from using 8, instead of 4, for the radius of the circular base.

▼ 2.7 — D

Choice D is correct. The volume, V, of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V, of a hemisphere is given by $V = \frac{1}{2}\frac{4}{3}\pi r^3$, or $V = \frac{2}{3}\pi r^3$. Substituting 27 for r in this formula yields

$V = \frac{2}{3}\pi 27^3$, which gives $V = 13{,}122\pi$, or V is approximately equal to $41{,}223.98$. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is $41{,}200$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.8 — A

Choice A is correct. The volume, V, of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of the base of the cylinder and h is the height. It's given that a right circular cylinder has a volume of 432 cubic centimeters and the area of the base is 24 square centimeters. Substituting 432 for V and 24 for πr^2 in the formula $V = \pi r^2 h$ yields $432 = 24h$. Dividing both sides of this equation by 24 yields $18 = h$. Therefore, the height of the cylinder, in centimeters, is 18.

Choice B is incorrect. This is the area of the base, in square centimeters, not the height, in centimeters, of the cylinder.

Choice C is incorrect. This is the height, in centimeters, of a cylinder if its volume is 432 cubic centimeters and the area of its base is 2, not 24, cubic centimeters.

Choice D is incorrect. This is the height, in centimeters, of a cylinder if its volume is 432 cubic centimeters and the area of its base is $\frac{1}{24}$, not 24, cubic centimeters.

▼ 2.9 — C

Choice C is correct. The two diagonals of square ABCD divide the square into 4 congruent right triangles, where each triangle has a vertex at the origin of the graph shown. The formula for the area of a triangle is $A = \frac{1}{2}bh$, where b is the base length of the triangle and h is the height of the triangle. Each of the 4 congruent right triangles has a height of 5 units and a base length of 5 units. Therefore, the area of each triangle is $A = \frac{1}{2}(5)(5)$, or 12.5 square units. Since the 4 right triangles are congruent, the area of each is $\frac{1}{4}$ of the area of square ABCD. It follows that the area of the square ABCD is equal to 4×12.5, or 50 square units.

Choices A and D are incorrect and may result from using 5 or 25, respectively, as the area of one of the 4 congruent right triangles formed by diagonals of square ABCD. However, the area of these triangles is 12.5. Choice B is incorrect and may result from using 5 as the length of one side of square ABCD. However, the length of a side of square ABCD is $5\sqrt{2}$.

▼ 2.10 — D

Choice D is correct. The volume, V, of a sphere can be found using the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. It's given that the sphere has a radius of $\frac{17}{5}$ feet. Substituting $\frac{17}{5}$ for r in the formula $V = \frac{4}{3}\pi r^3$ yields $V = \frac{4}{3}\pi\frac{17}{5}^3$, which is equivalent to $V = \frac{4}{3}\pi\frac{4,913}{125}$, or $V = \frac{19,652\pi}{375}$. Therefore, the volume, in cubic feet, of the sphere is $\frac{19,652\pi}{375}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the volume, in cubic feet, of a sphere with a radius of $\sqrt[3]{\frac{17}{5}}$ feet.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.11 — D

Choice D is correct. The volume, V, of a right square prism is given by the formula $V = s^2 h$, where s represents the length of the edge of the base and h represents the height of the prism. It's given that the volume of a right square prism is $2{,}880$ cubic units and the length of the edge of the base is 6 units. Substituting $2{,}880$ for V and 6 for s in the formula $V = s^2 h$ yields $2{,}880 = 6^2 h$, or $2{,}880 = 36h$. Dividing both sides of this equation by 36 yields $80 = h$. Therefore, the height, in units, of the prism is 80.

Choice A is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is 6 units and the volume of the prism is $144\sqrt{30}$, not $2{,}880$, units.

Choice B is incorrect. This is the area, in square units, of the base, not the height, in units, of the prism.

Choice C is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is 6 units and the volume of the prism is $864\sqrt{5}$, not $2{,}880$, units.

▼ 2.12 — B

Choice B is correct. It's given that rectangles $ABCD$ and $QRST$ are similar, where A, B, C, and D correspond to Q, R, S, and T, respectively. It follows that $\overline{AB}$ corresponds to $\overline{QR}$ and $\overline{AD}$ corresponds to $\overline{QT}$. If two rectangles are similar, then the lengths of their corresponding

sides are proportional. It's given in the figure that the length of $\bar{AB}$ is 12 inches, the length of $\bar{QR}$ is 5 inches, and the length of $\bar{QT}$ is 10 inches. If x is the length, in inches, of $\bar{AD}$, then $\frac{12}{5}$ is equivalent to $\frac{x}{10}$. Therefore, the value of x can be found using the equation $\frac{12}{5} = \frac{x}{10}$. Multiplying each side of this equation by 10 yields $\frac{120}{5} = x$, or $24 = x$. Therefore, the length, in inches, of $\bar{AD}$ is 24.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length, in inches, of $\bar{QT}$, not $\bar{AD}$.

▼ 2.13 — D

Choice D is correct. The area, A, of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the circle has a radius of 43 meters. Substituting 43 for r in the formula $A = \pi r^2$ yields $A = \pi 43^2$, or $A = 1{,}849\pi$. Therefore, the area, in square meters, of the circle is $1{,}849\pi$.

Choice A is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{\frac{43}{2}}$ meters.

Choice B is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{43}$ meters.

Choice C is incorrect. This is the circumference, in meters, of the circle.

▼ 2.14 — D

Choice D is correct. The volume, V, of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of 41 inches. Substituting 41 inches for s in this equation yields $V = 41^3$ cubic inches, or $V = 68{,}921$ cubic inches. Therefore, the volume of the cube is $68{,}921$ cubic inches.

Choice A is incorrect. This is the perimeter, in inches, of the cube.

Choice B is incorrect. This is the area, in square inches, of a face of the cube.

Choice C is incorrect. This is the surface area, in square inches, of the cube.

▼ 2.15 — D

Choice D is correct. It's given that triangle XYZ is similar to triangle TUV. Therefore, each side of triangle XYZ is k times its corresponding side of triangle TUV, where k is a constant. It follows that the perimeter of triangle XYZ is k times the perimeter of triangle TUV. It's also given that $\bar{TU}$ corresponds to $\bar{XY}$ and the length of $\bar{TU}$ is 18. Let x represent the length of $\bar{XY}$. It follows that $x = 18k$. The table shows that the perimeters of triangles TUV and XYZ are 37 and 333, respectively. It follows that $333 = 37k$, or $9 = k$. Substituting 9 for k in the equation $x = 18k$ yields $x = 189$, or $x = 162$. Therefore, the length of $\bar{XY}$ is 162.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\bar{TU}$, not the length of $\bar{XY}$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.16 — 14

The correct answer is 14. The volume, V, of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that the right circular cylinder has a height of 16 inches and a volume of $3{,}136\pi$ cubic inches. Substituting 16 for h and $3{,}136\pi$ for V in the formula $V = \pi r^2 h$ yields $3{,}136\pi = \pi r^2 (16)$, or $3{,}136\pi = 16\pi r^2$. Dividing both sides of this equation by 16π yields $196 = r^2$. Since r represents the radius of the cylinder, the value of r is positive. Taking the positive square root of both sides of this equation yields $r = 14$. Therefore, the radius, in inches, of the cylinder is 14.

▼ 2.17 — ?

The correct answer is 360. The volume of a right rectangular prism is calculated by multiplying its dimensions: length, width, and height. Multiplying the values given for these dimensions yields a volume of $(4)(9)(10) = 360$ cubic centimeters.

▼ 2.18 — A

Choice A is correct. The volume of a right circular cylinder with a radius of r is the product of the area of the base, πr^2, and the height, h. The volume of the right circular cylinder described is 45π and its height is 5. If the radius is r, it follows that $45\pi = \pi(r)^2(5)$. Dividing both sides of this equation by 5π yields $9 = r^2$. Taking the square root of both sides yields $r = 3$ or $r = -3$. Since r represents the radius, the value must be positive. Therefore, the radius is 3.

Choice B is incorrect and may result from finding that the square of the radius is 9, but then from dividing 9 by 2, rather than taking the square root of 9. Choice C is incorrect. This represents the square of the radius. Choice D is incorrect and may result from solving the equation $45\pi = \pi(r)^2(5)$ for r^2, not r, by dividing by π on both sides and then by subtracting, not dividing, 5 from both sides.

▼ 2.19 — C

Choice C is correct. It's given that triangle *TUV* is similar to triangle *XYZ*, and $\bar{TU}$ corresponds to $\bar{XY}$. If two triangles are similar, then the ratio of their perimeters is equal to the ratio of their corresponding sides. It's given that the perimeter of triangle *TUV* is 50, the perimeter of triangle *XYZ* is 150, and the length of $\bar{TU}$ is 6. Let *n* represent the length of $\bar{XY}$. It follows that $\frac{50}{150} = \frac{6}{n}$, or $\frac{1}{3} = \frac{6}{n}$. Multiplying each side of this equation by *n* yields $\frac{n}{3} = 6$. Multiplying each side of this equation by 3 yields $n = 18$. Therefore, the length of $\bar{XY}$ is 18.

Choice A is incorrect. This is the solution to $\frac{3}{1} = \frac{6}{n}$, not $\frac{1}{3} = \frac{6}{n}$.

Choice B is incorrect. This is the length of $\bar{TU}$, not $\bar{XY}$.

Choice D is incorrect. This is the sum of the length of $\bar{TU}$ and the perimeter of triangle *TUV*, not the length of $\bar{XY}$.

▼ 2.20 — D

Choice D is correct. The volume, *V*, of a right circular cone is given by the formula $V = \frac{1}{3}\pi r^2 h$, where πr^2 is the area of the circular base of the cone and *h* is the height. It's given that this right circular cone has a volume of $71{,}148\pi$ cubic centimeters and the area of its base is $5{,}929\pi$ square centimeters. Substituting $71{,}148\pi$ for *V* and $5{,}929\pi$ for πr^2 in the formula $V = \frac{1}{3}\pi r^2 h$ yields $71{,}148\pi = \frac{1}{3}5{,}929\pi h$. Dividing each side of this equation by $5{,}929\pi$ yields $12 = \frac{h}{3}$. Multiplying each side of this equation by 3 yields $36 = h$. Let *s* represent the slant height, in centimeters, of this cone. A right triangle is formed by the radius, *r*, height, *h*, and slant height, *s*, of this cone, where *r* and *h* are the legs of the triangle and *s* is the hypotenuse. Using the Pythagorean theorem, the equation $r^2 + h^2 = s^2$ represents this relationship. Because $5{,}929\pi$ is the area of the base and the area of the base is πr^2, it follows that $5{,}929\pi = \pi r^2$. Dividing both sides of this equation by π yields $5{,}929 = r^2$. Substituting $5{,}929$ for r^2 and 36 for *h* in the equation $r^2 + h^2 = s^2$ yields $5{,}929 + 36^2 = s^2$, which is equivalent to $5{,}929 + 1{,}296 = s^2$, or $7{,}225 = s^2$. Taking the positive square root of

both sides of this equation yields $85 = s$. Therefore, the slant height of the cone is 85 centimeters.

Choice A is incorrect. This is one-third of the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice B is incorrect. This is the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice C is incorrect. This is the radius, in centimeters, of the base, not the slant height, in centimeters, of this cone.

▼ 2.21 — D

Choice D is correct. It's given that rectangle A and rectangle B are similar. Therefore, there is a positive constant k such that the length of each side of rectangle B is k times the length of the corresponding side of rectangle A and the area of rectangle B is k^2 times the area of rectangle A. It's also given that the area of rectangle A is 630 square inches and that the area of rectangle B is $2{,}520$ square inches. It follows that $2{,}520 = 630k^2$. Dividing both sides of this equation by 630 yields $4 = k^2$, or $k^2 = 4$. Taking the positive square root of each side of this equation yields $k = 2$. Therefore, the length of each side of rectangle B is 2 times the length of the corresponding side of rectangle A. It follows that the perimeter of rectangle B is 2 times the perimeter of rectangle A. Since the perimeter of rectangle A is 210 inches, the perimeter of rectangle B is $2(210)$, or 420, inches. Therefore, $n = 420$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from using the area scale factor instead of the side length scale factor.

▼ 2.22 — 2592/5, 518.4

The correct answer is 518.4. It's given that the area of the original poster is 360 square inches. Let l represent the length, in inches, of the original poster, and let w represent the width, in inches, of the original poster. Since the area of a rectangle is equal to its length times its width, it follows that $360 = lw$. It's also given that a copy of the poster is made in which the length and width of the original poster are each increased by 20%. It follows that the length of the copy is the length of the original poster plus 20% of the length of the

original poster, which is equivalent to $l + \frac{20}{100}l$ inches. This length can be rewritten as $l + 0.2l$ inches, or $1.2l$ inches. Similarly, the width of the copy is the width of the original poster plus 20% of the width of the original poster, which is equivalent to $w + \frac{20}{100}w$ inches. This width can be rewritten as $w + 0.2w$ inches, or $1.2w$ inches. Since the area of a rectangle is equal to its length times its width, it follows that the area, in square inches, of the copy is equal to $1.2l1.2w$, which can be rewritten as $1.21.2lw$. Since $360 = lw$, the area, in square inches, of the copy can be found by substituting 360 for lw in the expression $1.21.2lw$, which yields $1.21.2360$, or 518.4. Therefore, the area of the copy, in square inches, is 518.4.

▼ 2.23 — D

Choice D is correct. The area of a rectangle is given by bh, where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle $ABCD$, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy. Therefore, the area, in square units, of $ABCD$ can be represented by the expression xy. It's given that the length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. Therefore, the length, in units, of the base of $EFGH$ can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $6x6y$, which is equivalent to $36xy$. Therefore, the area, in square units, of $EFGH$ can be represented by the expression $36xy$. It's given that the area of $ABCD$ is 54 square units. Since xy represents the area, in square units, of $ABCD$, substituting 54 for xy in the expression $36xy$ yields 3654, or $1{,}944$. Therefore, the area, in square units, of $EFGH$ is $1{,}944$.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not 6, times the length of the corresponding side of $ABCD$.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{7}{3}}$, not 6, times the length of the corresponding side of $ABCD$.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not 6, times the length of the corresponding side of $ABCD$.

▼ 2.24 — 2216

The correct answer is $2{,}216$. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of 28 cm, a width of 15 cm, and a height of 16 cm. Thus, for this prism, there are two faces with area 2815 cm^2, two faces with area

2816 cm^2, and two faces with area 1516 cm^2. Therefore, the surface area, in cm^2, of the right rectangular prism is $22815 + 22816 + 21516$, or $2{,}216$.

▼ 2.25 — A

Choice A is correct. The total surface area of a right square pyramid is the sum of the combined surface area of its four lateral faces and the area of its square base. It's given that the right square pyramid has a total surface area of $70{,}560$ square inches, and the combined surface area of the four lateral faces of this pyramid is $38{,}160$ square inches. It follows that the area of the square base is the difference between the total surface area and the combined surface area of the four lateral faces, or $(70{,}560 - 38{,}160)$ square inches, which is equal to $32{,}400$ square inches. The area, B, of the base of a square pyramid is given by the equation $B = s^2$, where s is the side length of the base. Substituting $32{,}400$ for B in this equation yields $32{,}400 = s^2$. Taking the positive square root of both sides of this equation yields $180 = s$. The four lateral faces of a square pyramid are congruent triangles. It follows that the area of each lateral face is $\frac{38{,}160}{4}$, or $9{,}540$, square inches. The area, A, of each lateral face of a square pyramid is given by the equation $A = \frac{1}{2}sl$, where s is the side length of the pyramid's base and l is the pyramid's slant height. Substituting $9{,}540$ for A and 180 for s in this equation yields $9{,}540 = \frac{1}{2}(180)l$, or $9{,}540 = 90l$. Dividing both sides of this equation by 90 yields $106 = l$. In the right square pyramid, a right triangle can be drawn, where the lengths of the legs correspond to the pyramid's height, h, and half the side length of its base, $\frac{1}{2}s$, and the length of the hypotenuse corresponds to the pyramid's slant height, l. It follows that the relationship between h, s, and l is given by the equation $h^2 + (\frac{1}{2}s)^2 = l^2$. Substituting 180 for s and 106 for l in this equation yields $h^2 + ((\frac{1}{2})(180))^2 = (106)^2$, or $h^2 + 8{,}100 = 11{,}236$. Subtracting $8{,}100$ from both sides of this equation yields $h^2 = 3{,}136$. Taking the positive square root of both sides of the equation yields $h = 56$. Therefore, the height, in inches, of this pyramid is 56.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pyramid's slant height, not its height, in inches.

Choice D is incorrect. This is the side length of the pyramid's base, not its height, in inches.

▼ 2.26 — A

Choice A is correct. It's given that the length of each side of a scale model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of a ballroom. Therefore, the area of the scale model is $\frac{1}{10}^2$, or $\frac{1}{100}$, times the area of the actual floor of the ballroom. It's given that

the area of the floor of the ballroom is 600 square meters. Therefore, the area, in square meters, of the scale model is $\frac{1}{100}600$, or 6.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.27 — 50

The correct answer is 50. The area of a rectangle is lw, where l and w are the lengths of two adjacent sides of the rectangle. It's given that the points $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$ form a rectangle. The points $(-4, -1)$ and $(4, 5)$ form a side of the rectangle that is adjacent to the side of the rectangle formed by the points $(4, 5)$ and $(7, 1)$. In the xy-plane, the distance between two points, (x_1, y_1) and (x_2, y_2), can be found as $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Thus, the length, in units, of the side formed by the points $(-4, -1)$ and $(4, 5)$ can be found as $\sqrt{4 - (-4)^2 + 5 - (-1)^2}$, which can be simplified to $\sqrt{100}$, or 10. The length, in units, of the side formed by the points $(4, 5)$ and $(7, 1)$ can be found as $\sqrt{7 - 4^2 + 1 - 5^2}$, which can be simplified to $\sqrt{25}$, or 5. Therefore, the area, in square units, of the rectangle formed by the four points shown is $(10)(5)$, or 50.

▼ 2.28 — B

Choice B is correct. It's given that the base area of the right rectangular prism is $24t \text{ cm}^2$ and the length of the base is $\frac{8}{3} \text{ cm}$. Dividing the base area by the length yields the width: $\frac{24t}{\frac{8}{3}}$, or $(24t)\frac{3}{8}$, or $9t \text{ cm}$. It's also given that the height of the prism is 15 cm. A right rectangular prism has two rectangular bases and four rectangular lateral faces. The total area of the two bases is $2(24t)$, or $48t \text{ cm}^2$. The four lateral faces include two with dimensions $\frac{8}{3} \text{ cm}$ by 15 cm and two with dimensions $9t \text{ cm}$ by 15 cm. The total area of these four lateral faces is $2\frac{8}{3}(15) + 2(9t)(15) \text{ cm}^2$, which is equivalent to $80 + 270t \text{ cm}^2$. Adding this to the total area of the two bases yields $48t + 270t + 80$, or $318t + 80 \text{ cm}^2$. Thus, the expression, which represents the surface area, in cm^2, of the right rectangular prism is $318t + 80$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the volume, in cm^3, not the surface area, in cm^2.

▼ 2.29 — 4

The correct answer is 4. It's given that right rectangular prism P is similar to right rectangular prism Q. Therefore, there is a positive constant k such that the length of each edge of prism P is k times the length of the corresponding edge of prism Q and the volume of prism P is k^3 times the volume of prism Q. It's given that the volume of prism P is 972 cm^3 and the volume of prism Q is 36 cm^3. It follows that $972 = 36k^3$. Dividing both sides of this equation by 36 yields $\frac{972}{36} = k^3$, or $k^3 = 27$. Taking the cube root of both sides of this equation yields $k = 3$. It follows that the length of each edge of prism P is 3 times the length of the corresponding edge of prism Q. It's given that one edge of prism P has a length of 12 cm. It follows that 12 cm is 3 times the length of the corresponding edge of prism Q. Thus, the length, in cm, of the edge of prism Q corresponding to the edge of prism P with a length of 12 cm is $\frac{12}{3}$, or 4.

▼ 2.30 — C

Choice C is correct. The volume of right circular cylinder A is given by the expression $\pi r^2 h$, where r is the radius of its circular base and h is its height. The volume of a cylinder with twice the radius and half the height of cylinder A is given by $\pi(2r)^2\left(\frac{1}{2}\right)h$, which is equivalent to $4\pi r^2\left(\frac{1}{2}\right)h = 2\pi r^2 h$. Therefore, the volume is twice the volume of cylinder A, or $2 \times 22 = 44$.

Choice A is incorrect and likely results from not multiplying the radius of cylinder A by 2. Choice B is incorrect and likely results from not squaring the 2 in 2r when applying the volume formula. Choice D is incorrect and likely results from a conceptual error.

▼ 2.31 — 1260

The correct answer is 1,260. Since it's given that prisms X and Y are similar, all the linear measurements of prism Y are k times the respective linear measurements of prism X, where k is a positive constant. Therefore, the surface area of prism Y is k^2 times the surface area of prism X and the volume of prism Y is k^3 times the volume of prism X. It's given that the surface area of prism Y is $1{,}450 \text{ cm}^2$, and the surface area of prism X is 58 cm^2, which implies that $1{,}450 = 58k^2$. Dividing both sides of this equation by 58 yields $\frac{1{,}450}{58} = k^2$, or $k^2 = 25$. Since k is a positive constant, $k = 5$. It's given that the volume of prism Y is

$1{,}250 \text{ cm}^3$. Therefore, the volume of prism X is equal to $\frac{1{,}250}{k^3} \text{ cm}^3$, which is equivalent to $\frac{1{,}250}{5^3} \text{ cm}^3$, or 10 cm^3. Thus, the sum of the volumes, in cm^3, of the two prisms is $1{,}250 + 10$, or $1{,}260$.

▼ 2.32 — 594

The correct answer is 594. It's given that the right circular cone has a height of 22 cm and a base with a diameter of 18 cm. Since the diameter of the base of the cone is 18 cm, the radius of the base is 9 cm. The volume V, in cm^3, of a right circular cone can be found using the formula $V = \frac{1}{3} \pi r^2 h$, where h is the height, in cm, of the cone and r is the radius, in cm, of the base of the cone. Substituting 22 for h and 9 for r in this formula yields $V = \frac{1}{3} \pi 9^2 (22)$, or $V = 594 \pi$. Therefore, the volume of the cone is $594 \pi \text{ cm}^3$. It's given that the volume of the cone is $n \pi \text{ cm}^3$. Therefore, the value of n is 594.

▼ 2.33 — B

Choice B is correct. The surface area of a cube with side length s is equal to $6s^2$. Since the surface area is given as 54 square meters, the equation $54 = 6s^2$ can be used to solve for s. Dividing both sides of the equation by 6 yields $9 = s^2$. Taking the square root of both sides of this equation yields $3 = s$ and $-3 = s$. Since the side length of a cube must be a positive value, $s = -3$ can be discarded as a possible solution, leaving $s = 3$. The volume of a cube with side length s is equal to s^3. Therefore, the volume of this cube, in cubic meters, is 3^3, or 27.

Choices A, C, and D are incorrect and may result from calculation errors.

▼ 2.34 — 135/8, 16.87, 16.88

The correct answer is $\frac{135}{8}$. It's given that triangles ABC and DEF are similar and each side length of triangle ABC is 4 times the corresponding side length of triangle DEF. For two similar triangles, if each side length of the first triangle is k times the corresponding side length of the second triangle, then the area of the first triangle is k^2 times the area of the second triangle. Therefore, the area of triangle ABC is 4^2, or 16, times the area of triangle DEF. It's given that the area of triangle ABC is 270 square inches. Let a represent the area, in square inches, of triangle DEF. It follows that 270 is 16 times a, or $270 = 16a$. Dividing both sides of this equation by 16 yields $\frac{270}{16} = a$, which is equivalent to $\frac{135}{8} = a$. Thus, the

area, in square inches, of triangle DEF is $\frac{135}{8}$. Note that 135/8, 16.87, and 16.88 are examples of ways to enter a correct answer.

▼ 2.35 — C

Choice C is correct. It's given that rectangle *ABCD* is similar to rectangle *EFGH*. Therefore, if the length of each side of rectangle *ABCD* is *k* times the length of the corresponding side of rectangle *EFGH*, then the area of rectangle *ABCD* is k^2 times the area of rectangle *EFGH*. It's given that the area of rectangle *ABCD* is 648 square inches and the area of rectangle *EFGH* is 72 square inches. It follows that $k^2 = \frac{648}{72}$, or $k^2 = 9$. Taking the square root of each side of this equation yields $k = \sqrt{9}$, or $k = 3$. It follows that the length of each side of rectangle *ABCD* is 3 times the length of the corresponding side of rectangle *EFGH*. It's given that the length of the longest side of rectangle *ABCD* is 36 inches. Therefore, 36 inches is 3 times the length of the longest side of rectangle *EFGH*, and the longest side of rectangle *EFGH* is equal to $\frac{36}{3}$, or 12, inches.

Choice A is incorrect. This is the length, in inches, of the longest side of a rectangle with side lengths that are $\frac{1}{9}$ the corresponding side lengths of rectangle *ABCD*, rather than a rectangle with an area that is $\frac{1}{9}$ the area of rectangle *ABCD*.

Choice B is incorrect. This is the factor by which the area of rectangle *ABCD* is larger than the area of rectangle *EFGH*, not the length, in inches, of the longest side of rectangle *EFGH*.

Choice D is incorrect. This is the length, in inches, of the longest side of rectangle *ABCD*, not rectangle *EFGH*.

▼ 2.36 — 66

The correct answer is 66. It's given that the right circular cone has a height of 22 centimeters cm and a base with a diameter of $6\ \text{cm}$. Since the diameter of the base of the cone is $6\ \text{cm}$, the radius of the base is $3\ \text{cm}$. The volume *V*, $\text{in}\ \text{cm}^3$, of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where *h* is the height, $\text{in}\ \text{cm}$, and *r* is the radius, $\text{in}\ \text{cm}$, of the base of the cone. Substituting 22 for *h* and 3 for *r* in this formula yields $V = \frac{1}{3}\pi 3^2 22$, or $V = 66\pi$. Therefore, the volume of the cone is $66\pi\ \text{cm}^3$. It's given that the volume of the cone is $n\pi\ \text{cm}^3$. Therefore, the value of *n* is 66.

▼ 2.37 — B

Choice B is correct. The length of a segment in the *xy*-plane can be found using the distance formula, $\sqrt{x_2 - x_1^2 + y_2 - y_1^2}$, where x_1, y_1 and x_2, y_2 are the endpoints of the segment. The segment shown has endpoints at $-6, 4$ and $3, 10$. Substituting $-6, 4$ and $3, 10$ for x_1, y_1 and x_2, y_2, respectively, in the distance formula yields $\sqrt{3 - -6^2 + 10 - 4^2}$, or $\sqrt{9^2 + 6^2}$, which is equivalent to $\sqrt{81 + 36}$, or $\sqrt{117}$. Let x represent the length, in units, of the other leg of this triangle. The area, A, of a right triangle can be calculated using the formula $A = \frac{1}{2}bh$, where b and h are the lengths of the legs of the triangle. It's given that the area of the triangle is $36\sqrt{13}$ square units. Substituting $36\sqrt{13}$ for A, $\sqrt{117}$ for b, and x for h in the formula $A = \frac{1}{2}bh$ yields $36\sqrt{13} = \frac{1}{2}\sqrt{117}x$. Multiplying both sides of this equation by 2 yields $72\sqrt{13} = x\sqrt{117}$. Dividing both sides of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{13}}{\sqrt{117}} = x$. Multiplying the numerator and denominator on the left-hand side of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{1{,}521}}{117} = x$, or $\frac{7239}{117} = x$, which is equivalent to $\frac{2{,}808}{117} = x$, or $24 = x$. Therefore, the length, in units, of the other leg of this triangle is 24.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $3\sqrt{13}$ is equivalent to $\sqrt{117}$, which is the length, in units, of the line segment shown in the *xy*-plane, not the length, in units, of the other leg of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.38 — B

Choice B is correct. The area, A, of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. It's given that the triangle is a right triangle. Therefore, its base and height can be represented by the two legs. It's also given that the triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. Since $\sqrt{80}$ units is the greatest of these lengths, it's the length of the hypotenuse. Therefore, the two legs have lengths $2\sqrt{2}$ and $6\sqrt{2}$ units. Substituting these values for b and h in the formula $A = \frac{1}{2}bh$ gives $A = \frac{1}{2}2\sqrt{2}6\sqrt{2}$, which is equivalent to $A = 6\sqrt{4}$ square units, or $A = 12$ square units.

Choice A is incorrect. This expression represents the perimeter, rather than the area, of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.39 — 12

The correct answer is 12. The volume, V, of a right square prism can be calculated using the formula $V = s^2 h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is 2,016 cubic units and the height is 14 units. Substituting 2,016 for V and 14 for h in the formula $V = s^2 h$ yields $2{,}016 = s^2 14$. Dividing both sides of this equation by 14 yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is 12.

▼ 2.40 — 1681

The correct answer is 1,681. It's given that trapezoid A and trapezoid B are similar. When two trapezoids are similar, the scale factor between their corresponding side lengths is k, and the scale factor between their areas is k^2. It's given that the length of each side of trapezoid A is 41 times the length of the corresponding side of trapezoid B, so the scale factor between their corresponding side lengths is 41. It follows that the scale factor between the areas is 41^2, or 1,681. Therefore, the area of trapezoid A is 1,681 times as large as the area of trapezoid B.

▼ 2.41 — C

Choice C is correct. It's given that parallelogram *ABCD* is similar to parallelogram *PQRS*. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k, the scale factor between their areas is k^2. It's given that the length of each side of parallelogram *PQRS* is 2 times the length of its corresponding side of parallelogram *ABCD*, so the scale factor between their corresponding side lengths is 2. It follows that the scale factor between their areas is 2^2, or 4. It's given that the area, in square centimeters, of parallelogram *ABCD* is 5. It follows that the area, in square centimeters, of parallelogram *PQRS* is $5(4)$, or 20.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.42 — A

Choice A is correct. The volume of a cube can be found by using the formula $V = s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V = 68^3$, or 314,432 cubic inches. The volume of a sphere can be found by using the

formula $V = \frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V = \frac{4}{3}\pi 34^3$, or approximately $164{,}636$ cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives $314{,}432 - 164{,}636 = 149{,}796$ cubic inches.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.43 — C

Choice C is correct. The volume of a right circular cylinder is equal to $\pi a^2 b$, where a is the radius of a base of the cylinder and b is the height of the cylinder. It's given that the cylinder shown has a radius of r and a height of h. It follows that the volume of the cylinder shown is equal to $\pi r^2 h$. It's given that the second right circular cylinder has a radius of R and a height of H. It follows that the volume of the second cylinder is equal to $\pi R^2 H$. Choice C gives $R = 7r$ and $H = 8h$. Substituting $7r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi 7r^2 8h$, or $\pi 49r^2 8h$, which is equivalent to $\pi 392r^2 h$, or $392\pi r^2 h$. This expression is equal to 392 times the volume of the cylinder shown, $\pi r^2 h$. Therefore, $R = 7r$ and $H = 8h$ could represent the radius R, in terms of r, and the height H, in terms of h, of the second cylinder.

Choice A is incorrect. Substituting $8r$ for R and $7h$ for H in the expression that represents the volume of the second cylinder yields $\pi 8r^2 7h$, or $\pi 64r^2 7h$, which is equivalent to $\pi 448r^2 h$, or $448\pi r^2 h$. This expression is equal to 448, not 392, times the volume of the cylinder shown.

Choice B is incorrect. Substituting $8r$ for R and $49h$ for H in the expression that represents the volume of the second cylinder yields $\pi 8r^2 49h$, or $\pi 64r^2 49h$, which is equivalent to $\pi 3{,}136r^2 h$, or $3{,}136\pi r^2 h$. This expression is equal to $3{,}136$, not 392, times the volume of the cylinder shown.

Choice D is incorrect. Substituting $49r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi 49r^2 8h$, or $\pi 2{,}401r^2 8h$, which is equivalent to $\pi 19{,}208r^2 h$, or $19{,}208\pi r^2 h$. This expression is equal to $19{,}208$, not 392, times the volume of the cylinder shown.

▼ 2.44 — D

Choice D is correct. The volume, V, of a right cylinder is given by the formula $V = \pi r^2 h$, where r represents the radius of the base of the cylinder and h represents the height. Since the height is 4 inches longer than the radius, the expression $r+4$ represents the height of each cylindrical container. It follows that the volume of each container is represented by the equation $V = \pi r^2(r+4)$. Distributing the expression πr^2 into each term in the parentheses yields $V = \pi r^3 + 4\pi r^2$.

Choice A is incorrect and may result from representing the height as $4r$ instead of $r+4$. Choice B is incorrect and may result from representing the height as $2r$ instead of $r+4$. Choice C is incorrect and may result from representing the volume of a right cylinder as $V = \pi rh$ instead of $V = \pi r^2 h$.

▼ 2.45 — 8

The correct answer is 8. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point $(6, 6)$ is halfway between the points $(6, 2)$ and $(6, 10)$, and is a distance of 4 units from each of those two points. The point $(6, 6)$ is also a distance of 4 units from $(2, 6)$. Because the point $(6, 6)$ is the same distance from all three points shown, it must be the center of the circle. Since that distance is 4, it follows that the radius of the circle is 4. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the given circle is $2\pi(4)$, or 8π. It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 8.

▼ 2.46 — 36504

The correct answer is 36,504. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of 474,552 cubic units. Substituting 474,552 for V in $V = s^3$ yields $474{,}552 = s^3$. Taking the cube root of both sides of this equation yields $78 = s$. Thus, the edge length of the cube is 78 units. Since each face of a cube is a square, it follows that each face has an edge length of 78 units. The area of a square can be found using the formula $A = s^2$. Substituting 78 for s in this formula yields $A = 78^2$, or $A = 6{,}084$. Therefore, the area of one face of this cube is 6,084 square units. Since a cube has 6 faces, the surface area, in square units, of this cube is 66,084, or 36,504.

▼ 2.47 — A

Choice A is correct. The equation for the volume of a right circular cone is $V = \frac{1}{3}\pi r^2 h$. It's given that the volume of the right circular cone is $\frac{1}{3}\pi$ cubic feet and the height is 9 feet. Substituting these values for V and h, respectively, gives $\frac{1}{3}\pi = \frac{1}{3}\pi r^2(9)$. Dividing both sides of the equation by $\frac{1}{3}\pi$ gives $1 = r^2(9)$. Dividing both sides of the equation by 9 gives $\frac{1}{9} = r^2$. Taking the square root of both sides results in two possible values for the radius, $\sqrt{\left(\frac{1}{9}\right)}$ or $-\sqrt{\left(\frac{1}{9}\right)}$. Since the radius can't have a negative value, that leaves $\sqrt{\left(\frac{1}{9}\right)}$ as the only possibility. Applying the quotient property of square roots, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, results in $r = \frac{\sqrt{1}}{\sqrt{9}}$, or $r = \frac{1}{3}$.

Choices B and C are incorrect and may result from incorrectly evaluating $\sqrt{\left(\frac{1}{9}\right)}$. Choice D is incorrect and may result from solving $r^2 = 9$ instead of $r^2 = \frac{1}{9}$.

▼ 2.48 — C

Choice C is correct. The volume, V, of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 meters. Therefore, $h = 6$. It's also given that the right circular cylinder has a base with a circumference of 20π meters. The circumference, C, of a circle is given by $C = 2\pi r$, where r is the radius of the circle. Substituting 20π for C in the formula $C = 2\pi r$ yields $20\pi = 2\pi r$. Dividing each side of this equation by 2π yields $10 = r$. Substituting 10 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi(10)^2 6$, or $V = 600\pi$. Therefore, the volume, in cubic meters, of the cylinder is 600π.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the lateral surface area, not the volume, of the cylinder.

Choice D is incorrect. This is the result of using the diameter, not the radius, for the value of r in the formula $V = \pi r^2 h$.

▼ 2.49 — 27556

The correct answer is $27{,}556$. The area of a square is s^2, where s is the side length of the square. Let x represent the length of each side of square B. Substituting x for s in s^2 yields x^2. It follows that the area of square B is x^2. It's given that square A has side lengths that are 166 times the side lengths of square B. Since x represents the length of each side of square B, the length of each side of square A can be represented by the expression $166x$. It follows that the area of square A is $166x^2$, or $27{,}556x^2$. It's given that the area of square A is k times the area of square B. Since the area of square A is equal to $27{,}556x^2$, and the area of square B is equal to x^2, an equation representing the given statement is $27{,}556x^2 = kx^2$. Since x represents the length of each side of square B, the value of x must be positive. Therefore, the value of x^2 is also positive, so it does not equal 0. Dividing by x^2 on both sides of the equation $27{,}556x^2 = kx^2$ yields $27{,}556 = k$. Therefore, the value of k is $27{,}556$.

▼ 2.50 — A

Choice A is correct. Since all cubes are similar to each other, cube X and cube Y are similar figures. In similar figures, if lengths differ by a scale factor of k, then areas differ by a scale factor of k^2. It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. It follows that the surface area, in square inches, of cube Y is $\left(\frac{3}{88}\right)^2$, or $\frac{9}{7{,}744}$, times the surface area, in square inches, of cube X. Therefore, the value of n is $\frac{9}{7{,}744}$.

Alternate approach: The surface area of a cube is $6s^2$, where s is the edge length. Let x represent the edge length, in inches, of cube X. Substituting x for s in $6s^2$ yields $6x^2$. Therefore, the surface area, in square inches, of cube X can be expressed as $6x^2$. It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. Therefore, the edge length, in inches, of cube Y can be expressed as $\frac{3}{88}x$, or $\frac{3x}{88}$. Substituting $\frac{3x}{88}$ for s in $6s^2$ yields $6\frac{3x}{88}^2$, which is equivalent to $6\frac{9x^2}{7{,}744}$, or $\frac{27x^2}{3{,}872}$. Therefore, the surface area, in square inches, of cube Y can be expressed as $\frac{27x^2}{3{,}872}$. It's given that the surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. It follows that $\frac{27x^2}{3{,}872} = n(6x^2)$. Dividing both sides of this equation by $6x^2$ yields $\frac{27x^2}{3{,}872(6x^2)} = n$, which is equivalent to $\frac{27x^2}{23{,}232x^2} = n$, or $\frac{9}{7{,}744} = n$. Therefore, the value of n is $\frac{9}{7{,}744}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.51 — 24.5, 49/2

The correct answer is 24.5. It's given that a triangle is formed by connecting the three points shown, which are $-3, 4$, $5, 3$, and $4, -3$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $-3, 4$, $5, 4$, $5, -3$, and $-3, -3$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $5, -3$. This distance is $4 - -3$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of the rectangle is 78, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $5, 4$, and $5, 3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 4$ and $5, 3$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}81$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $4, -3$, $5, 3$, and $5, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 3$ and $5, -3$. This distance is $3 - -3$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, -3$ and $4, -3$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}16$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $-3, -3$, and $4, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $4, -3$ and $-3, -3$. This distance is $4 - -3$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $-3, 4$ and $-3, -3$. This distance is $4 - -3$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}77$, or 24.5. Thus, the area, in square units, of the triangle formed by connecting the three points shown is $56 - 4 - 3 - 24.5$, or 24.5. Note that 24.5 and 49/2 are examples of ways to enter a correct answer.

▼ 2.52 — 5800

The correct answer is $5{,}800$. The volume, V, of a right prism with a square base is given by the formula $V = s^2 h$, where s is the side length of the square base and h is the height of the prism. It's also given that a right prism has a square base with side lengths of 10 inches. It's also given that the height of the prism is 58 inches. Substituting 10 for s and 58 for h in the

formula $V = s^2 h$ yields $V = (10)^2 (58)$, or $V = 5{,}800$. Therefore, the volume, in cubic inches, of the prism is $5{,}800$.

▼ 2.53 — ?

Choice D is correct. The surface area is found by summing the area of each face. A right rectangular prism consists of three pairs of congruent rectangles, so the surface area is found by multiplying the areas of three adjacent rectangles by 2 and adding these products. For this prism, the surface area is equal to $2(4 \cdot 5) + 2(5 \cdot 6) + 2(4 \cdot 6)$, or $2(20) + 2(30) + 2(24)$, which is equal to 148.

Choice A is incorrect. This is the area of one of the faces of the prism. Choice B is incorrect and may result from adding the areas of three adjacent rectangles without multiplying by 2. Choice C is incorrect. This is the volume, in cubic inches, of the prism.

▼ 2.54 — B

Choice B is correct. Let x represent the side length, in cm, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2 \ \text{cm}^2$. It's given that this resulting surface area is equal to $\frac{92}{47}K \ \text{cm}^2$, so $2K - 2x^2 = \frac{92}{47}K$. Subtracting $\frac{92}{47}K$ from both sides of this equation yields $2K - \frac{92}{47}K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47}K - \frac{92}{47}K - 2x^2 = 0$, or $\frac{2}{47}K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47}K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K, in cm^2, of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is $90 \ \text{cm}$ and the side length of each square base is $x \ \text{cm}$, it follows that the area of each square base is $x^2 \ \text{cm}^2$ and the area of each lateral face is $90x \ \text{cm}^2$. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 490x$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x45x - 360$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding 360 to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by 45 yields $x = 8$. Since a side length of a rectangular prism can't be 0, the length of each square base is $8 \ \text{cm}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.55 — D

Choice D is correct. The area of a circle can be found by using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi 3n^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi 129n^2$, or $16{,}641\pi n^2$. Dividing the area of circle B by the area of circle A gives $\frac{16{,}641\pi n^2}{9\pi n^2}$, which simplifies to $1{,}849$. Therefore, the area of circle B is $1{,}849$ times the area of circle A.

Choice A is incorrect. This is how many times greater the radius of circle B is than the radius of circle A.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the coefficient on the term that describes the radius of circle B.